

Kartik Chauhan

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EDUCATION

B.Tech in Computer Science, UPES, Dehradun CGPA: 7.54/10	Jul 2023 – Till Date
ISC (Class XII), St. Joseph’s Academy, Dehradun Percentage: 89.6%	Apr 2022 – Mar 2023
ICSE (Class X), St. Joseph’s Academy, Dehradun Percentage: 90.33%	Apr 2020 – Mar 2021

CERTIFICATIONS & ACHIEVEMENTS

Smart India Hackathon 2026 – Top 50 Teams, Internal Round, UPES Dehradun (Aug–Sep 2026)
JNCIA-Junos – Juniper Networks (HPE Networking Education Services) (Sep 2026)
Network Defense – Cisco Networking Academy, UPES (May 2026)
Internship Certificate: Survey of India, Dehradun (2025)
Internship Certificate: Janki Devi Educational Welfare Society (2024)

TECHNICAL SKILLS

Languages	C++, Java, Python
DSA & Problem Solving	350+ problems solved (200+ LeetCode) – Arrays, Trees, Graphs, DP, Sorting
Cybersecurity	Zero Trust Architecture, Network Reconnaissance, Vulnerability Assessment, Nmap, Wireshark, Netcat, Metasploit (basic), Ports & Protocols
Networking	TCP/IP, OSI Model, Firewalls, Network Traffic Analysis
Databases	MySQL
Tools & Platforms	Kali Linux, Ubuntu

EXPERIENCE

Survey of India | Intern – Geodetic & Research Branch, Dehradun July 2025

- Engineered an Automated Geomagnetic Data Processing System under the Geophysical Wing, reducing manual bulletin preparation effort significantly.
- Preprocessed and validated raw geomagnetic sensor data, ensuring data integrity for scientific analysis across multiple observatory stations.
- Delivered all assigned modules within the internship timeline, receiving official certification from a Government of India organization.

PROJECTS

UEBA Real-Time Anomaly Detection System GitHub

- Trained an Isolation Forest model (200 estimators, 5% contamination) on 11 behavioral features – failed logins, night logins, unique devices/files, session duration, bytes transferred – to detect insider threats.
- Built a pipeline: raw logs → per-user feature aggregation (pandas groupby) → StandardScaler → anomaly score mapped to a 0–100 risk score.
- Wrote a Windows Event Log collector pulling Security/System/Application events via PowerShell and converting them into the model’s input schema.
- Built a Streamlit dashboard that reads live or uploaded CSVs, re-scores on every refresh, and ranks users by risk score for analyst review.
- Validated the model on a synthetic 652-record dataset with 6 planted suspicious users – all 6 were correctly flagged.

Sentinel – Zero Trust IoT Security System GitHub

- Built a Zero Trust security system for IoT devices where no device is trusted by default – every request is verified via JWT token and device status before access is granted.
- Developed REST APIs using Flask for device registration, authentication, JWT-based authorization, sensor data transmission, and device blocking.
- Built an interactive web dashboard using HTML, CSS, and JavaScript to monitor all connected IoT devices and manage their access in real time.
- Wrote a Python device simulator that periodically sends authenticated temperature readings to the server, mimicking real IoT hardware behavior for testing.

Geomagnetic Bulletin Automation Suite GitHub

- Government scientists at Survey of India were manually creating magnetic field reports from raw sensor data every month – built a Python desktop app that does this automatically in minutes.
- The app reads raw sensor files (.SAB format), processes the numbers using Pandas and NumPy, and generates a clean monthly and yearly magnetic bulletin report ready to publish.
- Added a graph view using Matplotlib so scientists can visually see how the magnetic field changed over time, with missing data points filled in automatically.
- Wrapped everything in a simple Tkinter desktop window so scientists with no coding knowledge can run the whole thing by just clicking buttons – validated output matched official reports at 94–95% accuracy.